

Extension of the sub-channel code ANTEO+ to the mixed convection regime



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HIGHLIGHTS

- The code structure is presented.
- The performed validation is outlined.
- Results are critically discussed assessing code accuracy.
- Conclusions are drawn and ground for future work identified.

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ABSTRACT

Liquid Metal cooled Fast Reactors, and notably Heavy Liquid Metal cooled systems, are promising options for achieving, in a relatively short term, the various advantages brought about by Small Modular Reactors (SMRs). To increase safety, natural circulation in the primary circuit to cool the core, even at rated power, is also considered, possibly bringing the core thermal exchange dynamics inside the mixed convection regime. To address, at the outset of the design process, thermal-hydraulics aspects, suitable tools must be employed, like the sub-channel code ANTEO+. ANTEO+ was, however, specifically built for the forced convection regime, trying to maximize modeling efficiency inside that regime; to make it applicable also for these SMRs concepts, its validity domain has been extended to the mixed convection regime. In this work, such extension process is outlined, presenting the new code structure, including the solved set of equations and models in the new anticipated application domain. A thorough validation for sub-channel and clad outer temperatures is also presented, confirming the ability of ANTEO+ in reproducing experimental data, homogeneously, in its new anticipated validity domain, with a relatively high degree of accuracy when compared to a reference tool like COBRA-IV-I-MIT. The need for Heavy Liquid Metal-cooled experiments dedicated to validation for sub-channel tools has also been highlighted so to greatly enhance confidence in the final code accuracy assessment, ultimately easing its use in the design phase.

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1. Introduction

Small Modular Reactors (SMRs) are gaining considerable momentum, both in the thermal and fast reactors communities, thanks to the opportunities for capital costs reduction and shorter construction times; very small or micro reactors are also actively considered for remote regions (e.g. arctic) applications, where the greed for cost-effective, reliable energy sources is particularly high.

Focusing on the fast reactors sector, among the many concepts proposed, much interest is gravitating around the ones cooled by Liquid Metals (LMs) like sodium, Lead-Bismuth Eutectic (LBE) and lead, thanks to the inherent flexibility offered by these coolants (Waltar et al., 2013). To make the systems more safe, natural circulation in the primary circuit to cool the core, even at rated power, is often employed (Smith et al., 2008) exploiting the strong buoyancy tendencies of LMs (particularly the heaviest ones). While a direct link between the primary circuit convection and the in-core thermal exchange regimes cannot be established, because specifically depending on the envisaged core and plant layouts, generally, when the first relies on natural circulation the possibility

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of exiting the forced convection regime entering into the mixed one concretely exists for the second.

Following the Generation IV philosophy, safety is deeply embedded in the core design phase, implying that thermal-hydraulic conditions must be quantitatively assessed at the beginning of the aforementioned phase; this can be achieved, exploiting dedicated tools like ANTEO+ (Lodi, 2016). ANTEO+ is a Sub-Channel (SC) code with the main objectives of simplifying the problem description without penalizing accuracy and of having a clear and identifiable application domain, so to help the user in the results interpretation phase which is necessary in a rational core design process; in Lodi (2016), it was demonstrated to be a reliable tool for design applications with a degree of accuracy superior to codes of the COBRA family (Fricano, 2012) while ensuring a much easier overall structure considerably easing convergence and cutting computational costs. However, ANTEO+ was specifically constructed for the core steady-state single-phase forced convection regime, so that, in order to make it applicable also to SMR concepts, the validity range has therefore to be extended to the steady-state single-phase mixed convection regime.

In this paper, the logics behind the regime extension is outlined, following the general principles described in Lodi (2016), namely: after having selected the application domain, the most appropriate set of equations and models must be coherently derived.

A comprehensive description of the new ANTEO+ structure in the mixed convection regime is presented in Section 2, stressing rationales behind the main simplifications. Next, the code validation on SC and clad outer temperatures is discussed, comparing ANTEO+ also with other codes in order to better contextualize the targeted accuracy, meaning, in the present work, the ability to reproduce experimental data. Finally, conclusions are drawn and further desired improvements, both in the code and from an experimental point of view, suggested.

2. Code structure

As explained in Section 1, the first step in the validity domain extension, is to set up a coherent system of equations suited for the target mixed convection regime. To this aim, in order to better identify the framework inside which the equation analysis and subsequent model selection has to be performed, a parameter Y_{mix} can be defined as

$$Y_{mix} = \frac{Gr_q}{Re}, \quad (1)$$

where Gr_q is the Grashof's number based on the applied heat flux and Re is the Reynolds's number; following Iannello et al. (1988) and Suh et al. (1989a,b) it can, conservatively, be said that mixed convection extends up to $Y_{mix} < 100$: when this condition is not fulfilled, the bundle has entered the natural convection regime, meaning that the main bundle parameters (e.g. velocity and temperature) are completely dominated by buoyancy-related effects. More practically, this means that situations with a very low power-to-flow ratio, such as flow blockages and transients, will exceed the mentioned limit, falling outside ANTEO+ validity domain, coherently with the fact that ANTEO+ targets nominal steady-state conditions.

The proposed criteria is subject to considerable uncertainties because it does not take into account the effect that geometry – $\frac{p_w}{D_H}$, number of pins and heated length – seems to have on the onset of natural convection; indeed, for widely spaced pins arrangements the transition starts for a (much) different value of Y_{mix} (e.g. up to 2000) (Iannello et al., 1988).

Finally, the additional constraint of having a set of equations that smoothly reduces to the one previously solved by ANTEO+ in forced convection (Lodi, 2016) has to be enforced, in order to assure physical coherency.

2.1. Mass conservation

The mass conservation equation for a SC reads

$$\frac{d\dot{m}_i}{dz} = - \sum_{j=1}^J W_{ij}, \quad (2)$$

where $\dot{m}_i$ is the mass flow rate of the i -th SC, z is the axial coordinate, J is number of SCs adjacent to i and W_{ij} is the transverse mass flow rate per unit length between SCs i and j . The main implicit assumption is that W_{ij} is thermally driven; as discussed in Section 2.3, indeed, pressure is taken uniform on the bundle cross section.

2.2. Axial momentum conservation

The axial momentum conservation equation can be generally expressed as

$$\begin{aligned} \frac{d(P_i A_i)}{dz} = & - \frac{d(\dot{m}_i v_i)}{dz} - A_i \rho_i g - \frac{1}{2} \rho_i v_i^2 A_i \frac{f_i}{D_{Hi}} - A_i \Delta P_{form,i} \\ & - \sum_{j=1}^J W_{ij} v^* - \sum_{j=1}^J W_{ij}^M (v_i - v_j), \end{aligned} \quad (3)$$

where, with reference to the i th SC, P is the pressure in the SC, A is its flow area, v is the coolant axial speed, ρ is the coolant density, g is the gravitational acceleration, f is the Darcy friction factor with the walls surrounding SC, D_H is the hydraulic diameter defined as $D_H = 4A/p_w$ with p_w indicating the wetted perimeter, ΔP_{form} is the form pressure loss per unit length due to changes in flow direction or geometry like the ones in the presence of spacers or at the inlet and outlet of the fuel bundle, v^* is the effective speed transported by cross flow and W_{ij}^M is the effective mass exchange rate between SCs i and j for momentum transfer per unit length. The last term is labeled as *effective* because it is a fictitious mass flow defined in order to preserve the turbulent momentum exchange between SCs; in single phase, indeed, turbulent mixing does not result in a net exchange of mass, even if there is a net exchange of momentum and energy.

The main simplification introduced pertains the momentum transfer via diversion cross flow, which is collapsed to a single term W_i , sum of the individual neighboring SCs j ($W_i = \sum_{j=1}^J W_{ij}$). The axial momentum equation to solve can then be cast as

$$\begin{aligned} \frac{d(P_i A_i)}{dz} = & - \frac{d(\dot{m}_i v_i)}{dz} - A_i \rho_i g - \frac{1}{2} \rho_i v_i^2 A_i \frac{f_i}{D_{Hi}} - A_i \Delta P_{form,i} - V^* W_i \\ & - \sum_{j=1}^J W_{ij}^M (v_i - v_j), \end{aligned} \quad (4)$$

where V^* is theoretically defined as $\frac{\sum_{j=1}^J W_{ij} v^*}{\sum_{j=1}^J W_{ij}}$; it should therefore symbolize an average of the confining SCs v^* . In the present context, and assuming smooth velocity gradients, V^* is taken as v_i ; this choice simplifies convergence while not impairing the overall accuracy (see Section 3).

2.3. Transverse momentum conservation

The vehicle of the diversion flow is assumed to be a redistribution of thermal origin (i.e. buoyancy); for this reason, the same assumption of uniform pressure on the bundle cross section put forward in Section 2.1 is here utilized bringing to

$$P_i = P_j \quad \forall i, j. \quad (5)$$

2.4. Energy conservation

The energy equation has the following form

$$\frac{d(\dot{m}_i h_i)}{dz} = \chi_i - \sum_{j=1}^J W_{ij} h^* - \sum_{j=1}^J W_{ij}^H (h_i - h_j) - \sum_{j=1}^J \rho_{ij} \alpha_{ij} \left(\frac{S}{\delta} \right) (h_i - h_j), \quad (6)$$

where h_i is the enthalpy of the i -th SC, χ_i is the linear power discharged to SC i from the surrounding fuel pins, h^* and W_{ij}^H are the energy counterparts of v^* and W_{ij}^M , ρ_{ij} is the average coolant density of SCs i and j , α_{ij} is the average thermal diffusivity and δ is an effective mixing length usually taken as the ratio of η and the conduction shape factor κ (Cheng, 1984).

Similarly to the axial momentum, the energy Eq. (6) is simplified, in its diversion term, as

$$\frac{d(\dot{m}_i h_i)}{dz} = \chi_i - H^* W_i - \sum_{j=1}^J W_{ij}^H (h_i - h_j) - \sum_{j=1}^J \rho_{ij} \alpha_{ij} \left(\frac{S}{\delta} \right) (h_i - h_j), \quad (7)$$

where H^* is the energy analog of V^* . The same assumption performed on V^* is done on H^* which is taken equal to h_i .

Having disregarded the transverse momentum equation a flow split model is necessary to establish the inlet conditions of each SC (Lodi, 2016). Because in mixed convection, the mass flow can change in a SC due to thermally induced redistribution (i.e. W_i), the effect of the flow split model on the final temperatures results is bound to be less important than in forced convection.

It can also be noticed that, as requested, for W_i approaching zero, the set of equations just described, reduces to the one solved in forced convection (Lodi, 2016).

2.5. Method of solution

The system of equations for mixed convection is somewhat more difficult to solve than the forced one, because now, the energy and momentum equations are interlinked and must therefore be solved together. The idea suggested in Cheng (1984) is to solve the system having as unknown the density and velocity variations in each SC along with the bundle pressure drop. To achieve this, first the mass conservation equation must be substituted into the energy and momentum ones so to eliminate the term W_i in favor of $\dot{m}_i = \rho_i v_i A_i$. Finally, to eliminate the enthalpy in favor of the density, advantage is taken of the relation $\Delta h \approx R \Delta \rho$ where $R = \left(\frac{\partial h}{\partial \rho} \right)_p$.

After a forward finite differences approximation of the involved terms, the final system of equations can be expressed as

$$\begin{aligned} E_i \Delta \rho_i + F_i \Delta v_i + \Delta P &= G_i + MEX_i \quad \forall i = 1, N_{sub} \\ S_i \Delta \rho_i + T_i \Delta v_i &= \frac{\chi_i \Delta z}{A_i} + EEX_i \quad \forall i = 1, N_{sub}, \end{aligned} \quad (8)$$

where N_{sub} is the number of SCs, E_i, F_i, U_i and T_i are coupling terms between momentum and energy, G_i is the hydrostatic pressure loss term while MEX_i and EEX_i are exchange terms between SCs for momentum and energy respectively, which include turbulent mixing, conduction effects and wire/grid contributions. Presently, there are $2N_{sub}$ equations and $2N_{sub} + 1$ unknowns – the density and velocity variations for each SC plus the bundle pressure drop – and so another equation, represented by the overall mass conservation, must be added,

$$\sum_{i=1}^{N_{sub}} \Delta \dot{m}_i = \sum_{i=1}^{N_{sub}} A_i (\Delta \rho_i (v_{iz} + \Delta v_i) + \rho_{iz} v_i) = 0. \quad (9)$$

The above system of equations is non-linear because the coupling terms depend on $\Delta \rho_i$ and Δv_i which are the unknowns; an iteration scheme is therefore necessary. To ease convergence, phenomena are divided in two categories:

- the ones related to flow redistribution like $\Delta \rho$ and Δv and
- the ones not related to buoyancy like W^H and the conduction term.

The first category is treated implicitly and contributes to the non-linearity of the system being included in the coupling coefficients E_i, F_i, U_i and T_i , while elements of the second are treated explicitly and so approximated with their value at the previous iteration and thus contributing to G_i, MEX_i and EEX_i . This is acceptable because they are usually small compared to redistribution in mixed convection conditions and, at the same time, are characterized by a slower axial variation.

The adopted iteration procedure, depicted in Fig. 2, starts, after the necessary flow split calculation, with a first guess of the density and velocity variations (e.g. $\Delta \rho_i = 0$ and $\Delta v_i = 0$ or the value of the previous axial step); then, the system is solved with Greene's method¹ (Greene, 1980), and the newly calculated values are used to update the coupling coefficients. The solution is believed to be converged when a set of physical conditions is satisfied; these include: the global mass balance, the bundle pressure drop difference, the temperature (and so the energy balance) and velocity (and so the momentum balance); convergence is reached when all these errors are below some user defined value ϵ_i (with $i = m, P, T, v$). Generally, the most stringent criteria happen to be the temperature and velocity ones, with the global mass balance and pressure drops quickly converging, as visible in Fig. 1. This set of criteria is believed sufficient to provide physically sound solutions in spite of the problem non-linearity, particularly when the research domain starts close to the limiting case of forced convection; issues could be expected very far from this point but, they will fall outside the validity domain of ANTEO+ in mixed convection (see Section 3.1) and so not posing particular concerns.

Thanks to various optimization strategies – like under-relaxation in the case of slow convergence – it has been found that the above procedure converges very well in every flow condition investigated, assuring calculation times lower than a minute on a standard personal computer.

2.6. Clad outer temperature

Once the coolant temperature field at an axial elevation has been calculated, the outer cladding temperature T_{co} can be derived as

$$T_{co}(z) = T_b(z) + \frac{\chi_p(z)}{\pi D \alpha_p}, \quad (10)$$

where T_b is the bulk coolant temperature surrounding the pin under analysis, χ_p is the linear power of the pin, D is the pin diameter and α_p is the convection coefficient, calculated via the Nusselt's number.

2.7. Constitutive relations for balance equations

To close the system of equations given by (8)–(10), state equations, concerning the involved material properties, and constitutive relations are needed; the former ones, in particular, have been taken equal to the forced convection set, described in Lodi

¹ Due to the fact that Gauss elimination is not suitable for sparse matrices, like the one of the present problem, Greene devised a dedicated procedure based on the linear combination of matrix rows to transform the original in a diagonal one.

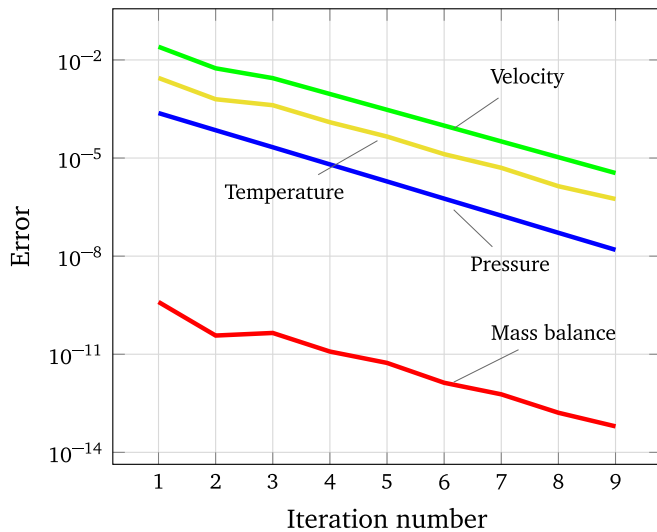


Fig. 1. Example of ANTEO+ convergence in a typical mixed convection situation.

(2016). Constitutive relations, however, are needed for f_i , $\Delta P_{form,i}$, W_{ij}^H , W_{ij}^M and δ . In this paper, only models for quantities that are different from their forced convection counterparts (the reader is directed to Lodi, 2016 for more information) or that are subject to constraints in the selected regime are discussed; they include f_i , W_{ij}^M and the conduction term.

2.7.1. Friction factor

At the beginning of Section 2 the parameter Y_{mix} was introduced; for $Y_{mix} > 100$, main bundle parameters are completely dominated by buoyancy related effects. The friction factor is one of such parameters; it has indeed been found that for increasing Y_{mix} the velocity field is more and more distorted, with mass flow pushed near the heated wall. Changing the velocity profile also changes the shear stress distribution and, ultimately, the friction factor (Suh et al., 1989a), that increases, introducing an error in the pressure drop estimation.

An additional problem arises in the evaluation of the gravity term of the momentum balance equations because, to properly assess it, the spatially-averaged coolant density should be used but, in lumped parameters tools like SC codes, only the bulk average is available. As Y_{mix} increases, the difference between the two densities can grow larger introducing an error in the hydrostatic pressure evaluation. To avoid this a corrective term should be added to the momentum equation or the friction factor suitably modified as suggested in Iannello et al. (1988).

Because, presently, ANTEO+ does not deal with the phenomena discussed in this section, calculation should be limited to $Y_{mix} < 100$, so to be confident that the error introduced by disregarding them is negligible.

2.7.2. Flow split

Due to the simplification of the transverse momentum equation a flow split model is needed exactly as explained in Lodi (2016). The main input of this model is the friction factor and so the flow split is subject to the same problems and restrictions that affect it; an important consideration must however be made and it starts observing that the flow split value of a SC does not depend on the absolute value of the friction factors but, rather, on their ratio, meaning that it is less sensitive to variations of the latter. High errors on f_i therefore do not necessarily imply high errors on X_i if in all the SCs the change of the friction factor is similar.

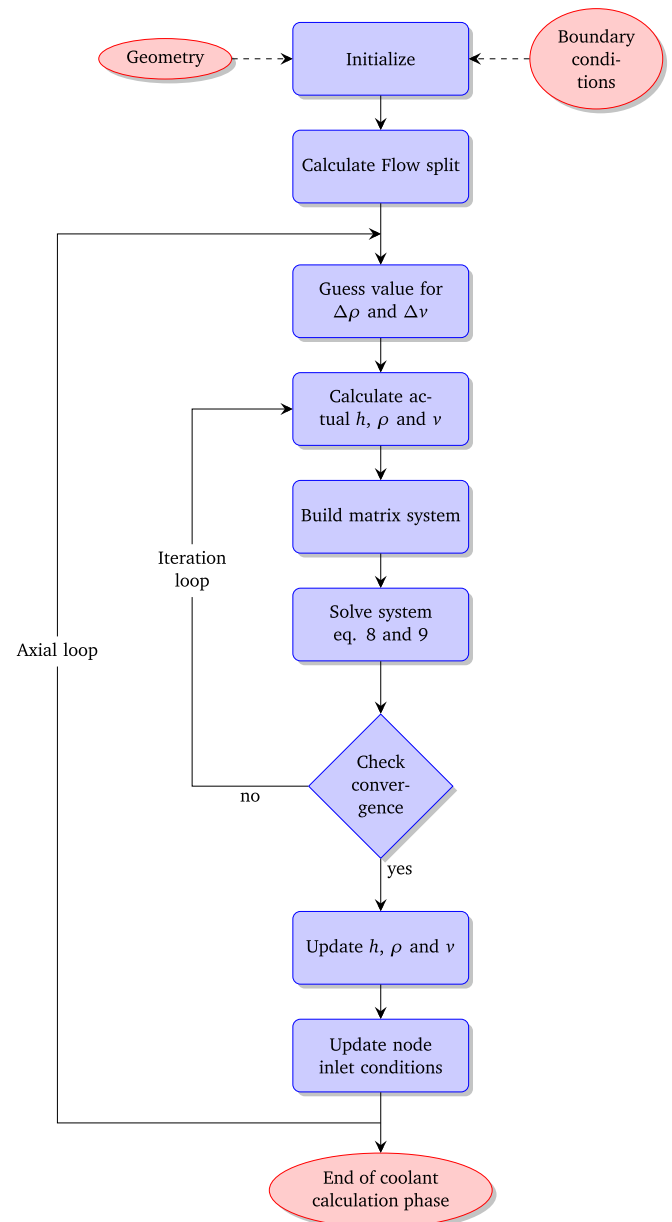


Fig. 2. Flow diagram of the coolant temperature calculation scheme in ANTEO+ for the mixed convection regime.

Moreover, due to flow redistribution, the temperature field is less sensitive to inlet conditions than in forced convection, further reducing the error on the SC temperature and so possibly aiding in keeping high ANTEO+ accuracy even near the boundary indicated by the Y_{mix} criterion.

2.7.3. Momentum mixing coefficient

The momentum mixing coefficient W_{ij}^M , representing the effective mass exchange rate between SCs i and j for momentum transfer per unit length, is taken equal to its energy counterpart W_{ij}^H described in Lodi (2016), because they are both driven by the same physical phenomena like turbulence, flow pulsation or wire sweeping.

2.7.4. Conduction

The main model modification, relatively to forced convection, pertains to the conduction term; while it is a minor contribution

in the forced regime, in mixed convection, indeed, azimuthal conduction through the rod can help energy exchange between SCs, especially if the heat flux at the outer rod surface is not azimuthally uniform. For an actual pin using oxides as fuel, the heat flux will be basically uniform at the inner cladding surface due to the low conductivity of the fuel and gap; at this point, coolant temperature non-uniformity around the pin – due for example to SC geometry variation in the azimuthal direction – can distort the heat flux distribution, making it azimuthally varying, due to the relatively high clad conductivity. This effect is bound to be magnified in tightly packed rod bundles.

In Ro and Todreas (1988) the authors suggest that the energy exchange via rod conduction is magnified, in many out-of-pile experimental setups, by the use of electric insulators like boron nitride featuring a thermal conductivity at least an order of magnitude higher than in the case of oxide, implying a stronger azimuthal gradient of the outer clad heat flux. To account for this phenomenon, a correlation, stemming from computer simulations on a multicell-multiregion layout cooled by sodium (and assuming a $\frac{k_{clad}}{k_{cool}}$ ratio around 0.26), was proposed having the following form

$$W_{ij}^C = 0.52 \left(\frac{k_p}{k_{cool}} \right)^{0.38} \left(\frac{P}{D} \right)^{-1.04} \left(\frac{S_{ij}}{D} \right)^{0.016} \rho_{ij} \alpha_{ij}, \quad (11)$$

where W_{ij}^C is the rod thermal conduction analog of W_{ij}^H , k_p is the equivalent internal pin structure conductivity – meaning inside the inner surface of the cladding – and k_{cool} is the coolant thermal conductivity. In Ro and Todreas (1988) k_p was selected equal to the boron nitride conductivity and specifically calibrated for its range of values; it is believed, however, that this correlation should be applicable also to real sodium-cooled fuel rods cases where the ratio $\frac{k_p}{k_{cool}}$ is quite low and the only contribution left is due to the cladding; indeed, boron nitride and the steel cladding typical of fast reactors have similar conductivities. Of course, for low k_p , the value of W_{ij}^C is bound to be small and thus not significantly contributing to the overall energy exchange dynamics, possibly making the proposed approximation acceptable.

Due to the Heavy LMs (HLMs) low thermal conductivity compared to sodium – around a factor of eight – the assumptions on which Eq. (11) is based do not longer hold:

- $\frac{k_p}{k_{cool}} < 0.4$ (in the case of boron nitride out-of-pile experiments) and
- $\frac{k_{clad}}{k_{cool}} \approx 0.26$.

The latter, in particular, makes the application of Eq. (11) to HLMs questionable; fortunately, a lower coolant conductivity means a lower overall impact of W_{ij}^C causing it to become negligible and, even more so, for real oxide fuel pins. Therefore, in ANTEO+, the present correlation is not applied to HLM-cooled cases.

2.7.5. Nusselt number

Similarly to the friction factor discussed in Section 2.7.1, the Nusselt's number is affected by the velocity distortion brought about by the increasing buoyancy (i.e. for increasing Y_{mix}). Again, however, with the shrewdness of the Y_{mix} criterion, this effect can be neglected and the Nusselt's number borrowed from forced convection (see Lodi, 2016; Lodi et al., 2017).

3. Validation

For the same motivations reviewed in Lodi (2016) validation is a critical part of a SC code development effort; since, in ANTEO+, the application range was decided beforehand (i.e. steady-state single-

phase mixed convection), during the validation stage, the additional task of verifying that the anticipated and actual validity domains coincide, so to support the derived set of equations and modeling approximations, arises.

Due to the code structure delineated in Section 2 some of the models, like the friction factor and flow split, have been borrowed from forced convection (Lodi, 2016) and so, they do not need specific validation in the mixed regime; indeed, pressure drop and flow split are usually experimentally measured in isothermal conditions and so, simulation results among the mixed and forced convection in ANTEO+ are bounded to be identical. The present section is therefore devoted to the validation of SC and outer clad temperatures.

3.1. SC temperature

The list of mixed convection simulated experiments, to be added to the ones presented in Lodi (2016) and Lodi et al. (2017), is reported in Table 1 (for compactness, not of all them will be directly presented). The validation will be presented separating forced and mixed convection and sodium and HLM coolants so to better highlight limitations and current experimental gaps.

The results will be presented highlighting the value of Y_{mix} , important in determining the degree of the velocity field distortion due to buoyancy, and thus its impact on the friction factor and Nusselt's number. As explained in Section 2, ANTEO+ regime in mixed convection extends until $Y_{mix} < 100$; to ease readiness the reported value will be normalized to 100 (i.e. $Y_{mix}^* \rightarrow \frac{Y_{mix}}{100}$) so that a $Y_{mix}^* > 1$ implies falling outside ANTEO+ validity domain.

Finally, it is mentioned that all the SC temperature calculations for wire-spaced bundles are performed with the correlation by Cheng and Todreas (1986) while, for gridded bundles, the mixing model by Kim and Chung (2001) has been used; the choice stems from their proven ability, in forced convection, of accurately reproducing experimental trends (see Lodi, 2016).

3.1.1. Forced convection test

For brevity, only a few selected cases, testing the ability of the code to consistently reproduce forced convection results will be presented, for both sodium- and HLM-cooled cases (the reader is referred to Lodi (2016) and Lodi et al. (2017) for the complete validation in forced convection).

Sodium-cooled experiment. In Fontana et al. (1973) a 19 pins bundle (here labeled as ORNL-19), part of the Fuel Failure Mockup facility, with the aim of testing various operating conditions of liquid metal cooled systems has been used. In particular, results for the forced convection Test 2 – Run 105 experiment, featuring an axially and radially uniform power profile, with a total power of 322.3 kW, a mass flow rate of 3.06 kg/s and a Y^* equal to 0.00, are reported in Fig. 3. Y^* , in analogy with Y_{mix} , is defined to discriminate between forced and mixed convection as $\frac{Gr}{Re^2}$ where Gr is the Grashof's number based on the axial temperature rise; from the results outlined in Connor and Carr (1978) the value of $2.5 \cdot 10^{-5}$ has been detected as the discriminant. Y^* is the normalized version of Y defined as $Y^* \rightarrow \frac{Y}{2.5 \cdot 10^{-5}}$, while the ordinate in Fig. 3 is the normalized temperature gain expressed as

$$\Delta T_{norm} = \frac{T - T_{in}}{T_b - T_{in}}, \quad (12)$$

where T_{in} is the inlet temperature and T_b is the average bundle outlet temperature.

² This definition of Y differs from the one in Lodi (2016) because it is believed to be more reliable and scalable to different coolants being inferred from more general grounds.

Table 1

Collection of experiments used for SC temperature validation in mixed convection, besides the one in Lodi (2016) and Lodi et al. (2017). Main physical and geometrical parameters are also given.

Identifier	Power (kW)	Power skew ^a	Flow rate (kg/s)	Y^*_{mix}
<i>Hexagonal – wire spacer</i>				
Morris et al. (1980), Na				
Run 12–144	12.3	Uniform	0.12	0.00
Run 17–105	51.4	1.43	0.23	0.00
Run 18–105	50.1	1.67	0.23	0.01
Run 19–105	49.4	2.39	0.23	0.01
<i>Engel et al. (1980), Na</i>				
Run 732	22.0	Uniform	0.18	0.00
Run 401	33.4	2.0	0.27	0.00
Run 720	14.2	2.0	0.13	0.00
Run 731	21.5	2.0	0.19	0.00
Run 731	21.5	2.0	0.19	0.00
<i>Lyu et al. (2016), LBE</i>				
NC	70.0	Uniform	5.6	0.22
<i>Hexagonal – grid spacer</i>				
Martelli et al. (2015), LBE				
Test 2 – NC	38.6	Uniform	9.0	10.8

^a Defined as the maximum to minimum ratio on the bundle cross section.

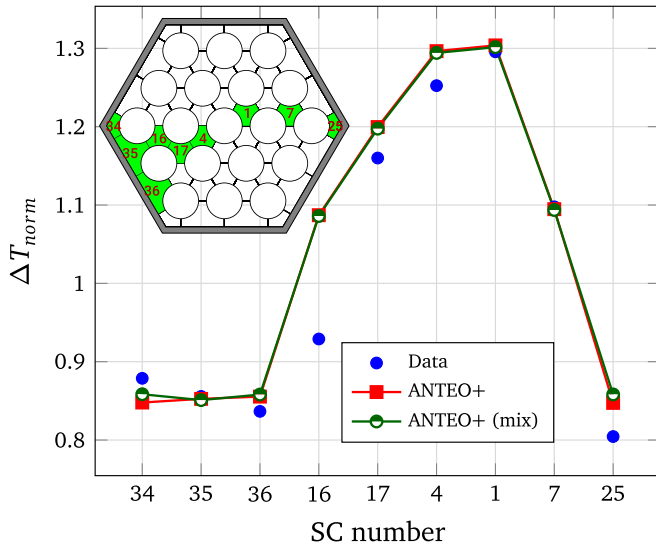


Fig. 3. Test 2 – Run 105 of the ORNL-19 bundle ($Y^* = 0.00$ and $Y^*_{mix} = 0.00$). Comparison of ANTEO+ and ANTEO+ for mixed convection (ANTEO+ (mix)) in a pure forced convection case. Instrumented SCs are also reported.

The average relative error $\bar{\epsilon}_{rel}$, defined as

$$\bar{\epsilon}_{rel} = \frac{1}{n} \sum_{i=1}^n \frac{|x_{exp} - x_{calc}|}{x_{exp}}, \quad (13)$$

with x_{exp} and x_{calc} indicating, respectively, the experimental and calculated values of the generic x parameter (e.g. ΔT_{norm}), for this run amounts to 3.9%.

Discussion. It is clearly visible from Fig. 3, how the set of equations and models presented in Section 2, smoothly reduces to one described in Lodi (2016) when the limit of the forced convection regime is encountered (i.e. $Y^* \rightarrow 0$).

The highest error is in the internal SC near the wall, possibly due to the inaccurate value of the mixing coefficient in this region; it must be said that being the power almost uniform in Fig. 3, SCs 7 and 16 should have similar temperatures, as predicted by the code, but not experimental data, giving an hint on experimental uncertainties in this bundle.

HLM-cooled experiment. An experimental campaign on a 19-pin electrically heated hexagonal bundle (see Fig. 4) with wire spacers cooled by LBE was conducted at the Karlsruhe Liquid metal Laboratory (KALLA) (Pacio et al., 2017) in the frame of the EU research project Safe ExploitAtion Related Chemistry for HLM reactors (SEARCH). The geometry and operating conditions were chosen so to be representative of the MYRRHA (Sarotto et al., 2013) fuel assembly.

The simulated case features an axially and radially uniform power profile, with a total power of 197 kW, a mass flow rate of 15.97 kg/s and a Y^* equating 0.06.

Results are presented in Fig. 4; the error $\bar{\epsilon}_{rel}$ amounts to 11.9%.

Discussion. Again, the results of ANTEO+ consistently reduces to those obtained by the original version of the code when the case falls into the forced convection limit. For what concerns the SC temperature distribution, it can be noted that ANTEO+ overpredicts the temperature for the interior SCs while, for the edge and corner ones, overestimates mixing. In the simulations, indeed, the edge and corner SCs have almost the same temperature while the data show that the corner one is colder. An important consideration must however be made: the difference between the SC center temperature (where the thermocouple is located) and the clad temperature oscillates between 10 °C and 25 °C which must be compared with a SC temperature increase around 100 °C³ – for the hottest SC at the location near the end of the heated length, and much lower for other SCs or elevations. This means that the SC center temperature is not representative of the bulk temperature, the quantity actually calculated by ANTEO+ and SC codes in general. This is partially proven by the good agreement between the predicted and experimental clad temperatures (see Fig. 9); error compensation between the SC temperature rise and the clad-coolant temperature difference predictions is a possibility but, alone, could not explain why the relative error drops from 11.9 to only 2.6 going from the SCs to the clad (see Section 3.2.1).

3.1.2. Mixed convection test

To highlight the ability of catching the buoyancy-driven redistribution phenomena, some significant sodium- and HLM-cooled experiments are presented and discussed.

³ As a comparison metric, the maximum value of the ratio between the bulk-clad temperature difference and the SC temperature rise found in all the sodium experiments analyzed, encompassing every SC and elevation, is 4.8%.

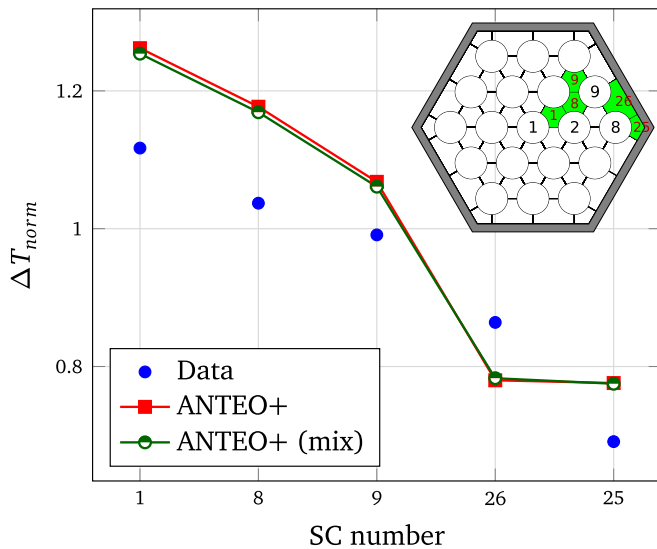


Fig. 4. KALLA-19 bundle ($Y^* = 0.06$ and $Y_{mix}^* = 0.00$). Comparison of ANTEO+ and ANTEO+ (mix) in a pure forced convection case. Instrumented SCs are also reported and data are taken at an elevation of 80.20 cm from the beginning of the heated section.

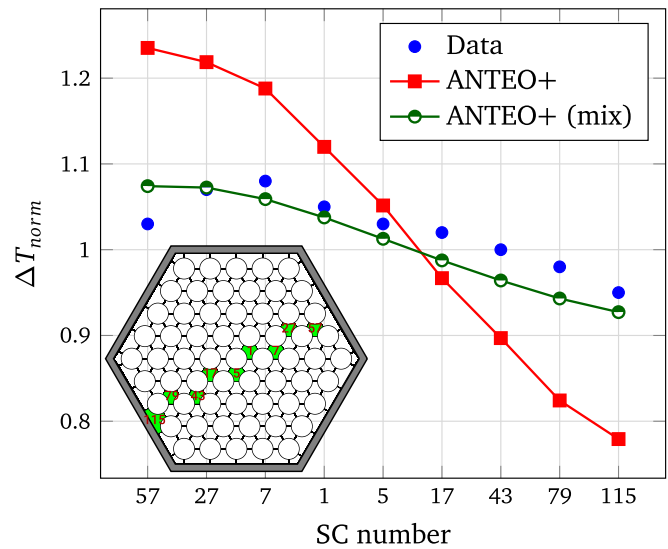


Fig. 5. Run 229 of the WARD-61 bundle ($Y^* = 49.1$ and $Y_{mix}^* = 0.00$). Comparison of ANTEO+ and ANTEO+ (mix) in a pure mixed convection case. Instrumented SCs are reported and data are taken at the end of the heated section.

Sodium-cooled experiment. The sodium-cooled mixed convection case with the highest Y^* (Lodi, 2016) but, with a Y_{mix}^* still much lower than unity, is reported in Fig. 5 comparing ANTEO+ for forced and mixed convection, where data are taken from the experimental bundle of the Westinghouse Advanced Reactors Division (here labeled as WARD-61) which has 61 electrically heated pins (Engel et al., 1980). In the WARD-61 bundle each row of pins has modular power and the axial power shape is a cosine with form factor of 1.4. In the Run229 test, a power skew of 2.8 was imposed with a total power of 17 kW, a mass flow rate of 0.129 kg/s and a Y^* equating 49.1.

Discussion. It can be seen the ability to properly cope with the strong flow redistribution in this operative conditions, enhanced by the strong radial power skew; the relative error on the ΔT_{norm} , indeed, drops from 10% – forced convection ANTEO+ – to 3.8% – mixed convection ANTEO+.

HLM-cooled experiment. Within the 6th EU Framework Program, ENEA assumed the commitment to perform an integral experiment aimed at simulating the primary flow path of a HLM-cooled – pool-type – nuclear reactor, implementing a new experimental activity named Integral Circulation Experiment (ICE) (Martelli et al., 2015) to be performed in the CIRCE facility. The ICE test section includes an assembly of 37 pins, spaced by grids, arranged in a hexagonal bundle (see Fig. 6a) with a wide pitch-to-diameter ratio of 1.8. Among the numerous transient tests, a mixed convection steady state run, labeled Test 2 – NC, was also performed featuring an axially and radially constant power profile with a total output of 38.6 kW, a mass flow rate of 9.0 kg/s and a Y_{mix}^* equal to 10.8, as reported in Table 1.

Results are reported in Fig. 6b; the average $\bar{\varepsilon}_{rel}$ amounts to 4.3%.

Discussion. Even if, in the reported operating conditions, Y_{mix}^* is greater than unity results are more than satisfactory; this could be due to the flat power profile helping in reducing buoyancy effects or to the conservativeness of the Y_{mix}^* criterion. As discussed in Section 2, indeed, for widely spaced bundles – like CIRCE-37 – the onset of pure free convection effects could be delayed. Further investigations are however necessary in order to reach a more reliable conclusion. Moreover, the difference between the two versions of ANTEO+ is relatively small for an $Y^* = 24.2$; this could be due to the flat power profile and the fact that only internal SCs were monitored.

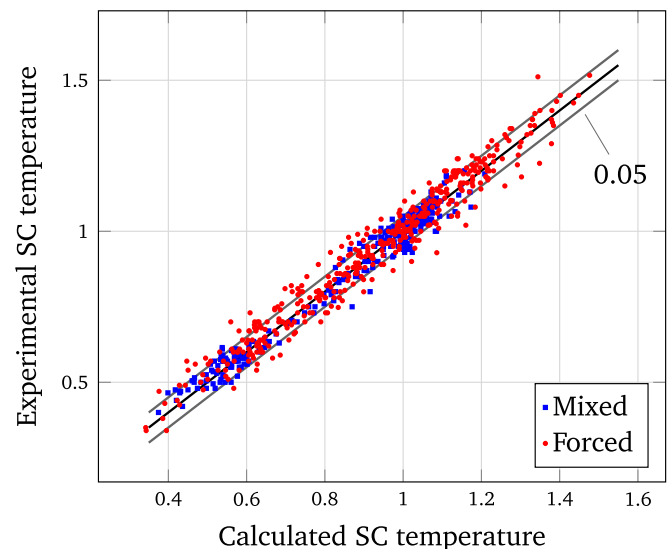


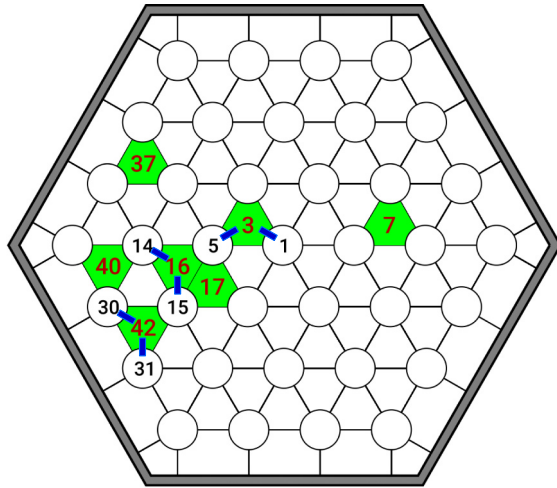
Fig. 6. Test 2 – NC of the CIRCE-37 bundle ($Y^* = 24.2$ and $Y_{mix}^* = 10.8$).

3.1.3. Results summary

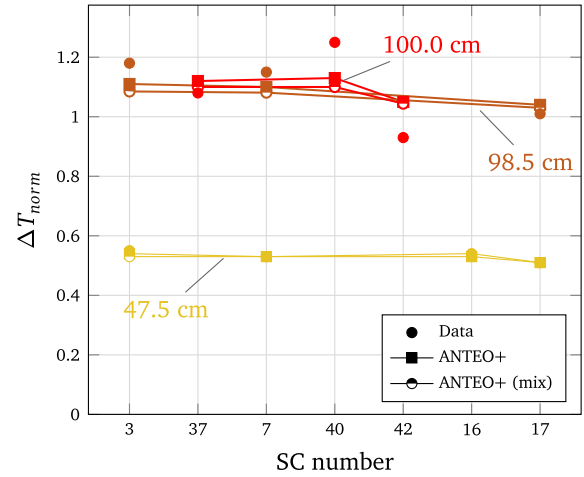
The overall validation results both in forced and mixed convection for sodium- and HLM-cooled experiments are now presented.

Sodium-cooled experiment. The overall results summary for the sodium-cooled cases of Table 1 and Lodi (2016) are displayed in Fig. 7 and quantitatively in Table 2 where a comparison between the forced and mixed convection ANTEO+ is also reported.

Discussion The ability of ANTEO+ to homogeneously cope with both forced and mixed convection is evident from Fig. 7 and Table 2; moreover, the direct comparison with COBRA-IV-MIT (Fricano, 2012) shows that ANTEO+ has an average $\bar{\varepsilon}_{rel}$ of 3.5 and 3.1% in the forced and mixed convection regimes respectively (taking into account only points where COBRA-IV-MIT results were available), while COBRA-IV-MIT assesses at 6.4 and 3.4% for the same parameter, well highlighting the advantages of the adopted modeling approach, with ANTEO+ able to maintain a uniform accuracy inside its validity domain, while COBRA-IV-MIT showing difficulties approaching the forced convection regime.



(a) Experimentally instrumented SCs and corresponding ANTEO+ numeration. Clad temperatures are measured and evaluated at the locations marked by the blue bars.



(b) Comparison of ANTEO+ and ANTEO+_{mix} in a mixed convection case. Elevations are measured from the end of the beginning of the heated section.

Fig. 7. Graphical results summary for the SC temperature validation of ANTEO+ in sodium-cooled bundles.

Table 2

ANTEO+ and ANTEO+ (mix) overall comparison in forced and mixed convection.

	$\bar{\epsilon}_{rel}[\%]$	90% confidence interval [%]
Sodium		
Forced convection (427 points)		
ANTEO+	3.8	8.6
ANTEO+ (mix)	3.9	8.6
Mixed convection (288 points)		
ANTEO+	4.9	11.7
ANTEO+ (mix)	2.9	5.7
HLMs		
Forced convection (20 points)		
ANTEO+	13.7	–
ANTEO+ (mix)	12.6	–
Mixed convection (20 points)		
ANTEO+	11.8	–
ANTEO+ (mix)	9.8	–

HLM-cooled experiment. A graphical summary of the SC temperature validation on HLMs is reported in Fig. 8 while a quantitative comparison between forced and mixed convection is reported in Table 2.

Discussion. While in forced convection the measured quantities are not representative of the SC bulk temperature, as previously discussed for the KALLA-19 bundle, in the mixed regime the difference between the two is bound to decrease; for this reason, the accuracy slightly increases, but still remains unsatisfactory low.

The main source of error, in the available data, comes from the KYLIN-61 bundle (Lyu et al., 2016) featuring a tight lattice pitch in combination with a wire spacer (see Lodi et al., 2017 for a more deep discussion); the correlations implemented in ANTEO+ presently seem to overestimate mixing, for near-wall SCs, in such HLM-cooled bundles, although energy balance errors may play an important role (Lodi, 2017). For widely spaced pin arrangements with grid spacers, the accuracy is more than satisfactory suggesting that Kim's correlation well scales from sodium to HLMs.

3.2. Clad outer temperature

Once the SC temperature is known the Nusselt's number must be calculated in order to compute the outer clad temperature, which is a tight design constraint (especially for HLM-cooled con-

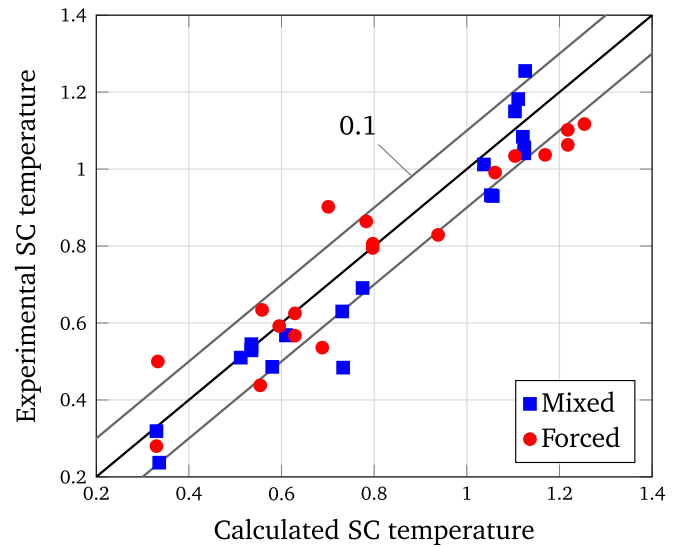


Fig. 8. Graphical results summary for the SC temperature validation of ANTEO+ in HLM-cooled bundles.

cepts). Some accuracy is therefore needed in order to reduce uncertainty margins and increase the degree of reliability of the overall core design.

Since no new sodium experiments are available, relatively to the forced convection validation presented in Lodi (2016) and given the proven ability of reproducing forced convection results, only HLM-cooled cases will be discussed. The list of forced and mixed convection experiments used in the HLM validation is reported in Table 3.

In this section, results for the clad outer temperature T_{co} are presented via the normalized quantity

$$\Delta T_{c,norm} = \frac{T_{co} - T_{in}}{T_b - T_{in}}, \quad (14)$$

thus merging together the SC temperature increase and the bulk-clad temperature rise, so to avoid the previously highlighted inconsistencies between the SC center and the SC bulk temperature.

Table 3

Collection of HLM-cooled experiments used for the clad outer temperature validation. Main physical and geometrical parameters are also given.

Ref.	Coolant	N_{pin}	P/D	H/D	Pe^a
Hexagonal – wire spacer					
Pacio et al. (2017)	LBE	19	1.279	40.0	1075
Lyu et al. (2016)	LBE	61	1.116	25.0	60–1050
Hexagonal – grid spacer					
Martelli et al. (2015)	LBE	37	1.800	–	345–2400

^a Peclet's number.

3.2.1. Forced convection test

This is the same bundle, with the same operating conditions presented in Section 3.1.1. Results are presented in Fig. 9, where the nomenclature is the following:

- “Mikityuk + Zhukov” indicates that the Nusselt's number has been calculated by means of the Mikityuk's correlation (Mikityuk, 2009) for the central pins and the Zhukov's correlation (IAEA, 1999) for the edge and corner pins;
- “Mikityuk” indicates that the Nusselt's number, for all the pins, has been calculated based on Mikityuk's correlation;
- “Ma” indicates that the Nusselt's number, for all the pins, has been calculated based on Ma's correlation (Ma et al., 2012).

For $\Delta T_{c,norm}$ the $\bar{\epsilon}_{rel}$ is 2.6, 3.4 and 3.4% for “Mikityuk + Zhukov”, “Mikityuk” and “Ma” respectively. Discussion As previously hinted, the error on $\Delta T_{c,norm}$ – clad – is much lower than the one on ΔT_{norm} – SC – partially confirming the hypothesis put forward in Section 3.1.1. From Fig. 9 it can also be seen how the correlations from “Mikityuk” and “Ma” perform similarly, both of them underpredicting the edge and corner pins temperatures. To increase accuracy, a dedicated correlation for these types of pin must be used, like the one from “Zhukov”.

3.2.2. Mixed convection test

Results for the CIRCE-37 bundle in the Test 2 – NC run previously presented are reported in Fig. 10 using Mikityuk's correlation for the Nusselt's number; error bars are inferred from the use of bifilar-type electrical pins as suggested in Martelli et al. (2015). The pin numeration is the same as Fig. 6a.

For $\Delta T_{c,norm}$, $\bar{\epsilon}_{rel}$ results for the “Mikityuk”⁴ and “Ma” correlations, assess at 8.0 and 8.5% respectively.

Discussion. The overall accuracy is satisfactory being the uncertainty, solely coming from the use of bifilar-type electrical pins (Martelli et al., 2015), on average, around 2.4% for the $\Delta T_{c,norm}$ and considering the Y_{mix}^* greater than unity.

It can also be noted that measurements at the grid plane (49.5 cm) are systematically higher than just upstream the spacer (47.5 cm) suggesting an influence of the latter when the coolant bulk-clad temperature difference is very small.

Finally, it is noticed that some of the measured clad temperatures are lower than the corresponding SC temperatures, implying local effects at the measuring position that are not captured, by definition, by SC codes; the representativeness of such points is therefore questionable.

3.2.3. Results summary

The summary of the clad temperature validation with the database in Table 3 is graphically reported in Fig. 11, where Mikityuk's correlation has been used, and quantitatively in Table 4.

⁴ The Zhukov's correlation is not used for the ICE test section because the Pe_D is out of the range of the correlation.

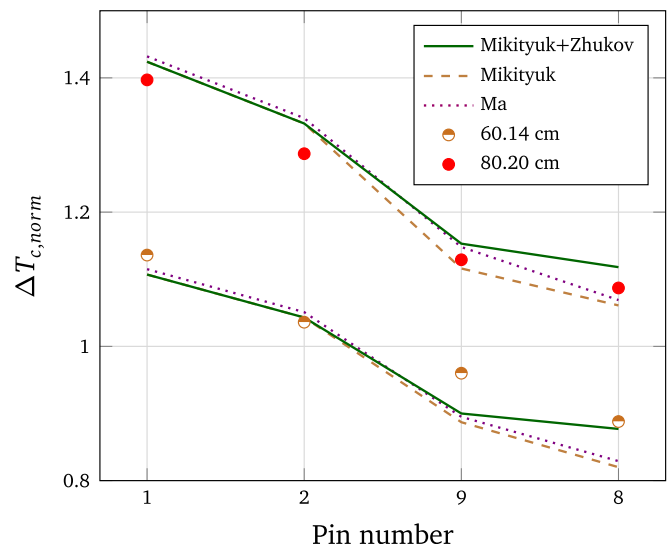


Fig. 9. Representation of experimental and ANTEO+ simulations clad temperatures for the KALLA-19 bundle. The pin identification number is explained in Fig. 4 and elevations are from the beginning of the heated section.

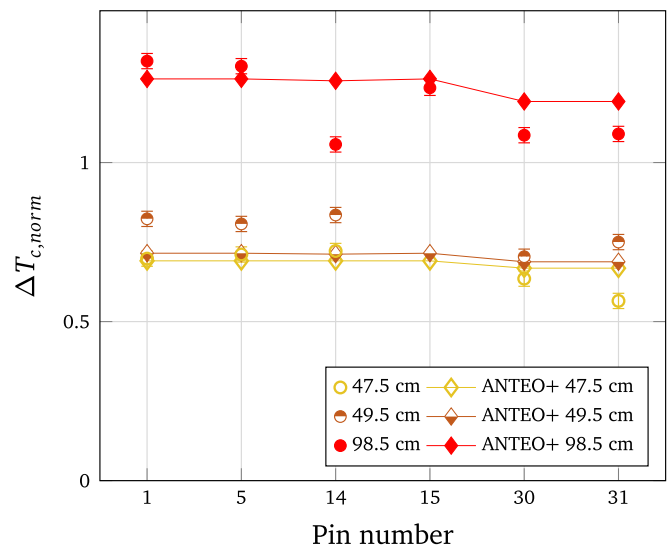


Fig. 10. Representation of ANTEO+ simulations and clad temperature experimental data for the CIRCE-37 bundle in mixed convection; reported elevations are from the beginning of the heated section.

Discussion. Again the ability to smoothly translate from forced to mixed convection is visible in Fig. 11, even with all the uncertainties previously discussed. The two Nusselt's correlations show similar performances, although KYLIN-61 (Lyu et al., 2016) mixed convection case falls outside Ma's application range.

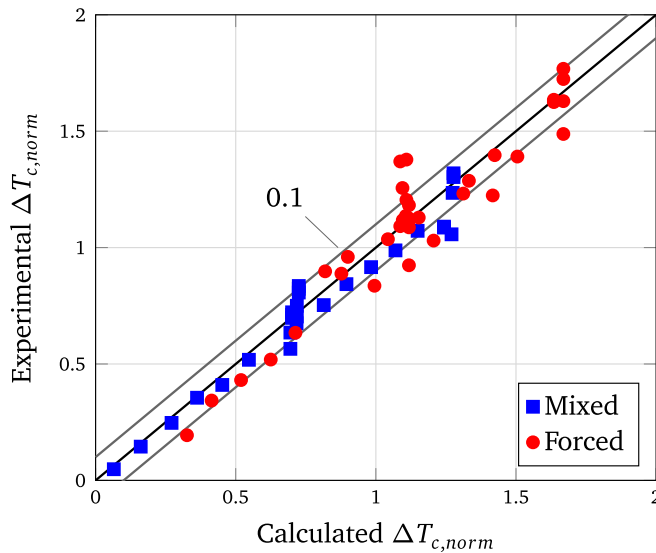


Fig. 11. Graphical results summary for the clad temperature validation of ANTEO+ in HLM-cooled bundles.

Table 4
ANTEO+ and ANTEO+ (mix) overall comparison in forced and mixed convection for the clad temperature predictions in HLM-cooled bundles.

			$\bar{\epsilon}_{rel}[\%]$
<i>Forced convection (36 points)</i>			
ANTEO	Mikityuk		9.8
	Ma		9.5
ANTEO (mix)	Mikityuk		9.8
	Ma		9.5
<i>Mixed convection (28 points)</i>			
ANTEO	Mikityuk		11.4
	Ma ^a		8.5
ANTEO (mix)	Mikityuk		8.7
	Maa		8.5

^a Do not include KYLIN-61 (Lyu et al., 2016) mixed convection data because outside correlation application range.

Finally it can be noticed how, comparing Tables 4 and 2, in mixed convection the errors on ΔT_{norm} and $\Delta T_{c,norm}$ are in line with each other, partially confirming the hypothesis put forward in forced convection about discrepancies between SC measured and actual bulk temperatures.

4. Validation summary

In the validation performed and discussed in the previous section, the ability of the modeling approach presented in Section 2 to smoothly translate from forced to mixed convection has been proven both in sodium- and HLM-cooled cases. Having borrowed flow split and pressure drops from the previous forced convection validation campaign (Lodi, 2016), attention has focused on SC and clad temperatures in mixed convection:

- SC temperatures are predicted with an average relative error of 2.9% for sodium-cooled bundles while, for HLM-cooled cases, accuracy decreases with a relative error up to 9.8%; excluding the KYLIN-61 bundle (Lyu et al., 2016), suspected of energy balance errors, the error assesses at 4.3%, much more in line with the sodium results.
- For the outer clad temperature, only HLM-cooled experiments have been presented, with an accuracy assessing around 8.7%, in line with sodium results presented in Lodi (2016) which, considering uncertainties and local effects, is deemed satisfactory.

Table 5

Summary of estimated uncertainties and confidence intervals of the main SC parameters in ANTEO+ in mixed convection. A reference experimental uncertainty is also given for qualitative comparison.

Parameters	$\bar{\epsilon}_{rel}[\%]$	Experimental uncertainty [%]
<i>Sodium</i>		
ΔT_{norm}	2.9	5
$\Delta T_{c,norm}$	–	–
<i>HLMs</i>		
ΔT_{norm}	4.3 ^a	5
$\Delta T_{c,norm}$	8.7	5.6 ^b

^a Excluding KYLIN-61 bundle mixed convection results.

^b Combination of the 5% uncertainty on the SC and 2.4% of bifilar-type simulator pins.

In Table 5 results are summarized and compared with the estimated experimental uncertainties in their measurement, so to better contextualize the validation results.

5. Conclusions

In the present work the SC code for thermal-hydraulic analysis of LM fast reactors, ANTEO+, initially conceived for steady-state single-phase forced convection applications, has been extended to the steady-state single-phase mixed convection regime.

The extension process has been pursued in line with the development methodology and philosophy originally outlined for ANTEO+, namely: after having selected the application domain, the most appropriate set of equations and models has been coherently derived so to maximize modeling efficiency.

The models and correlations implemented in the code, to support the aimed extension, have been presented stressing their expected validity domain. These expectations have been tested during the validation phase of the code, finding general agreement with experiments. In particular, a thorough validation has been performed on the SC temperatures and clad outer temperatures, quantifying the accuracy targeted by ANTEO+ for both sodium- and HLM-cooled bundles in mixed convection. Moreover, comparison with a reference SC code of the COBRA family has been performed revealing the general high level of ANTEO+ accuracy in its validity domain along with its ability to smoothly translate between forced and mixed convection without losing accuracy.

Work is still needed to extend the validation database in HLMs with particular emphasis on the SC temperatures in controlled conditions, along with measures of the clad temperatures avoiding local effects so to be truly representative of the quantity actually calculated by SC codes. Dedicated experiments to validation, indeed, will greatly enhance confidence in the final code accuracy assessment, easing its use in the design phase.

Concluding, ANTEO+ is here presented as a robust sub-channel code for application in single phase both in forced and mixed convection regimes, which enables the user to more easily follow the calculation process and thus facilitating the results interpretation phase, including the identification of the confidence interval of the results.

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References

- Cheng, S.K., 1984. Constitutive Correlations for wire-wrapped subchannel analysis under forced and mixed convection conditions (Ph.D. thesis). MIT.

- Cheng, S.K., Todreas, N.E., 1986. Hydrodynamic models and correlations for bare and wire-wrapped hexagonal rod bundles – bundle friction factors, subchannel friction factors and mixing parameters. *Nucl. Eng. Des.* 92, 227–251.
- Connor, M.A., Carr, A., 1978. Heat transfer in vertical tubes under conditions of mixed free and forced convection. In: *Proceeding of the 6th International Heat Transfer Conference*, Toronto, Canada.
- Engel, F.C. et al., 1980. Characterization of heat transfer and temperature distributions in an electrically heated model of an Imfbr blanket assembly. *Nucl. Eng. Des.* 62, 335–347.
- Fontana, M.H., et al., 1973. Temperature distribution in a 19-rod simulated Imfbr fuel assembly in a hexagonal duct (fuel failure mockup bundle 2a) – record of experimental data. Technical Report TM-4113, ORNL.
- Fricano, J.W., 2012. Integrated fuel performance and thermal-hydraulic subchannel models for analysis of sodium fast reactors (Ph.D. thesis). MIT.
- Greene, T., 1980. Development and evaluation of techniques for thermal-hydraulic analysis of LMFBFR fuel assemblies (Ph.D. thesis). MIT.
- IAEA, 1999. Imfr core and heat exchanger thermohydraulic design: Former ussr and present russian approaches. Technical Report TECDOC 1060, IAEA, Vienna.
- Iannello, V., Suh, K., Todreas, N., 1988. Mixed convection friction factors and nusselt numbers in vertical annular and subchannel geometries. *Int. J. Heat Mass Transfer* 31, 2175–2189.
- Kim, S., Chung, B.J., 2001. A scale analysis of the turbulent mixing rate for various prandtl number flow fields in rod bundles. *Nucl. Eng. Des.* 205, 281–294.
- Lodi, F., 2017. Development of Core Design Methods and Tools for Gen-IV Heavy Liquid Metal Cooled Reactors (Ph.D. thesis). UniBO.
- Lodi, F. et al., 2016. ANTEO+: a subchannel code for thermal-hydraulic analysis of liquid metal cooled systems. *Nucl. Eng. Des.* 301, 128–152.
- Lodi, F., et al., 2017. Extension to heavy liquid metals coolants of the validation database of the ANTEO+ sub-channel code. In: *Proceedings of the 3th International Conference on Fast Reactors and Related Fuel Cycles*, Yekaterinburg, Russia.
- Lyu, K. et al., 2016. Preliminary thermal-hydraulic sub-channel analysis of 61 wire-wrapped bundle cooled by lead bismuth eutectic. *Ann. Nucl. Energy* 92, 243–250.
- Ma, Z. et al., 2012. An innovative method for prediction of liquid metal heat transfer rate for rod bundles based on annuli. *Ann. Nucl. Energy* 47, 91–97.
- Martelli, D. et al., 2015. HLM fuel pin bundle experiments in the CIRCE pool facility. *Nucl. Eng. Des.* 292, 76–86.
- Mikityuk, K., 2009. Heat transfer to liquid metal: review of data and correlations for tube bundles. *Nucl. Eng. Des.* 234, 680–687.
- Morris, R.H., et al., 1980. Single phase sodium test in 61-pin full-length simulated LMFBFR assembly – record of phase 1 experimental data for THOR bundle 9. Technical Report TM-7313, ORNL.
- Pacio, J., et al., 2017. Thermal-hydraulic study of the LBE-cooled fuel assembly in the MYRRHA reactor: experiments and simulations. *Nucl. Eng. Des.* 312, 327–337.
- Ro, T.S., Todreas, N.E., 1988. Energy transfer mechanisms in Imr rod bundles under mixed convection conditions. *Nucl. Eng. Des.* 108, 343–357.
- Sarotto, M. et al., 2013. The MYRRHA-FASTEF cores design for critical and sub-critical operational modes (EU FP7 central design team project). *Nucl. Eng. Des.* 265, 184–200.
- Smith, C.F. et al., 2008. SSTAR: the US lead-cooled fast reactor (LFR). *J. Nucl. Mater.* 376, 255–259.
- Suh, K., Todreas, N., Rohsenow, W., 1989a. Mixed convective low flow pressure drop in vertical rod assemblies- i. predictive model and design correlation. *J. Heat Transfer* 111, 956–965.
- Suh, K., Todreas, N., Rohsenow, W., 1989b. Mixed convective low flow pressure drop in vertical rod assemblies- ii. Experimental validation. *J. Heat Transfer* 111, 966–973.
- Waltar, A.E., Tood, D.R., Tsvetkov, P.V., 2013. *Fast Spectrum Reactors*. Springer.